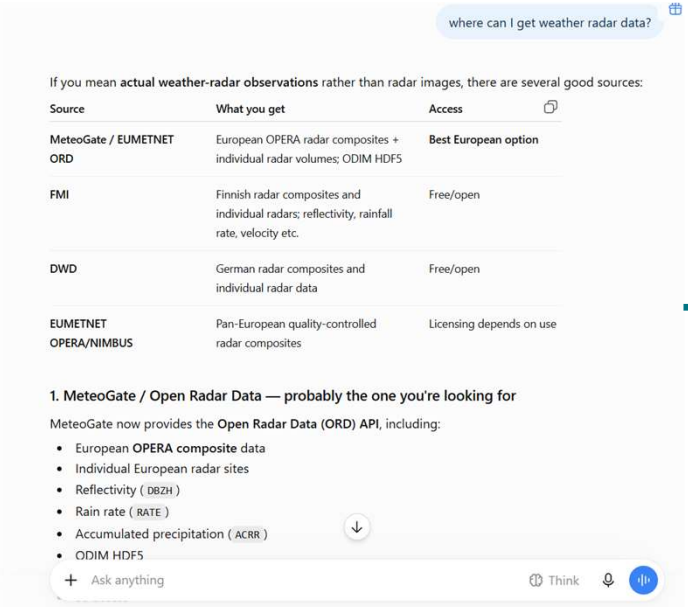
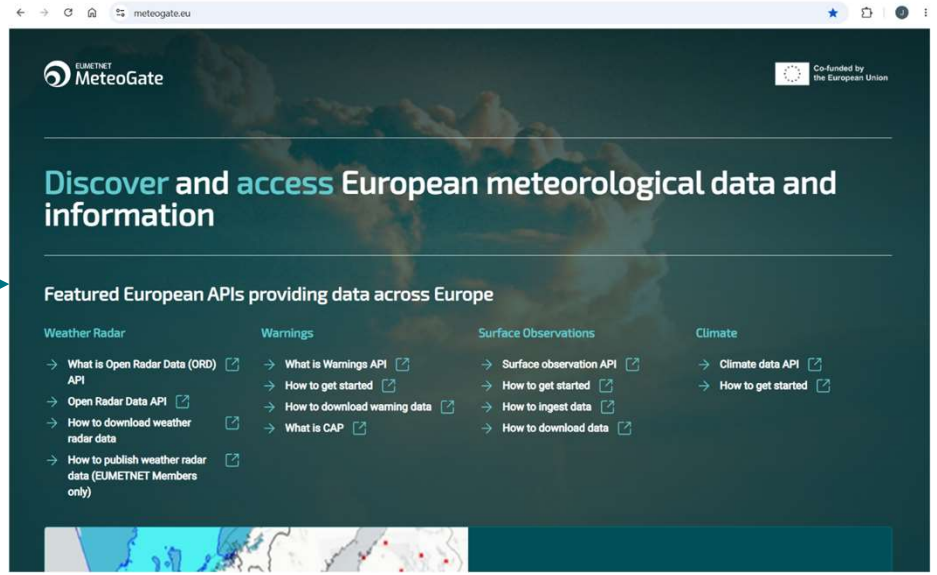
MeteoGate

A ONE-STOP SHOP FOR EUROPEAN WEATHER DATA

In Europe, EUMETNET (a network of 33 National Meteorological and Hydrological Services (NMHSs)) have developed MeteoGate.eu – a data space where you can discover and access their weather and climate data. So, when you ask e.g. ChatGPT where to find weather information, it sends you to their official data.

MeteoGate is an **interface** for users and NMS's data. It is not a database, it does not collect data. The data stays where it is. NMHSs continue to store their own data and keep control of their data licenses.





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MeteoGate

WEBSITE

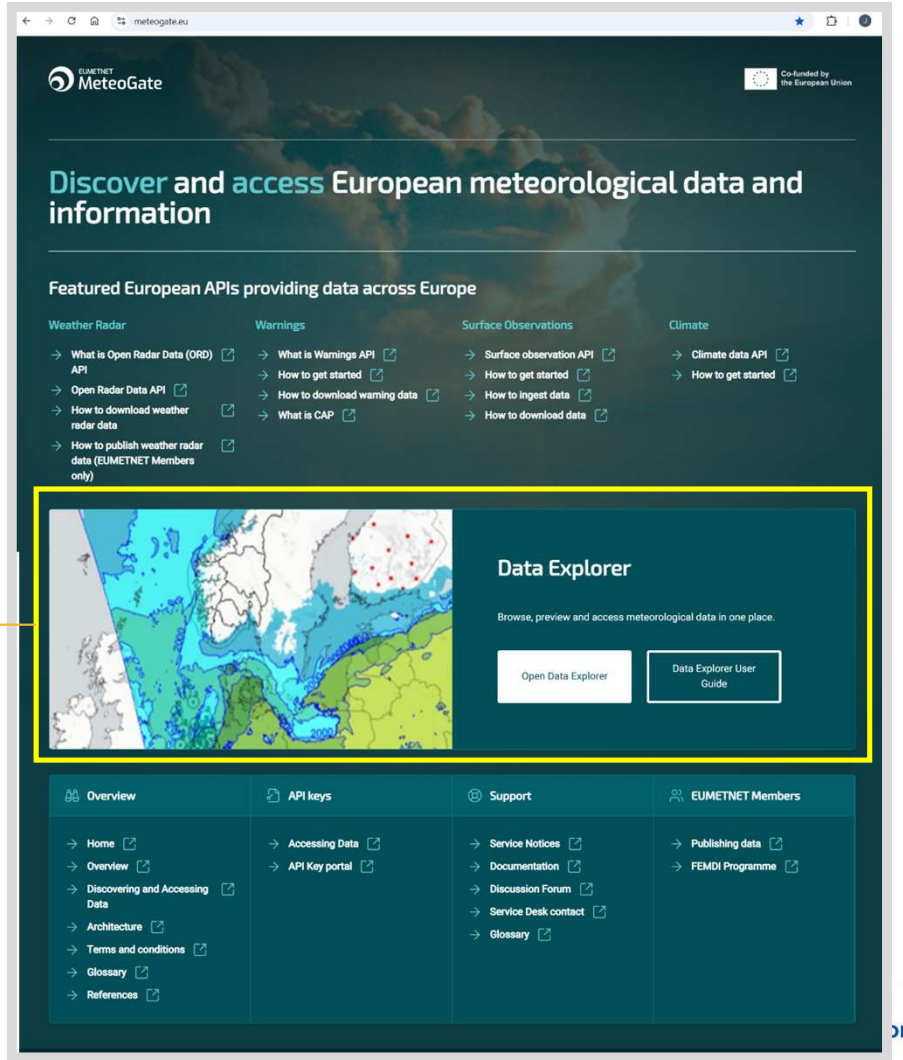
Everything needed is linked from the MeteoGate website.

DISCOVER DATA

MeteoGate's Data Explorer

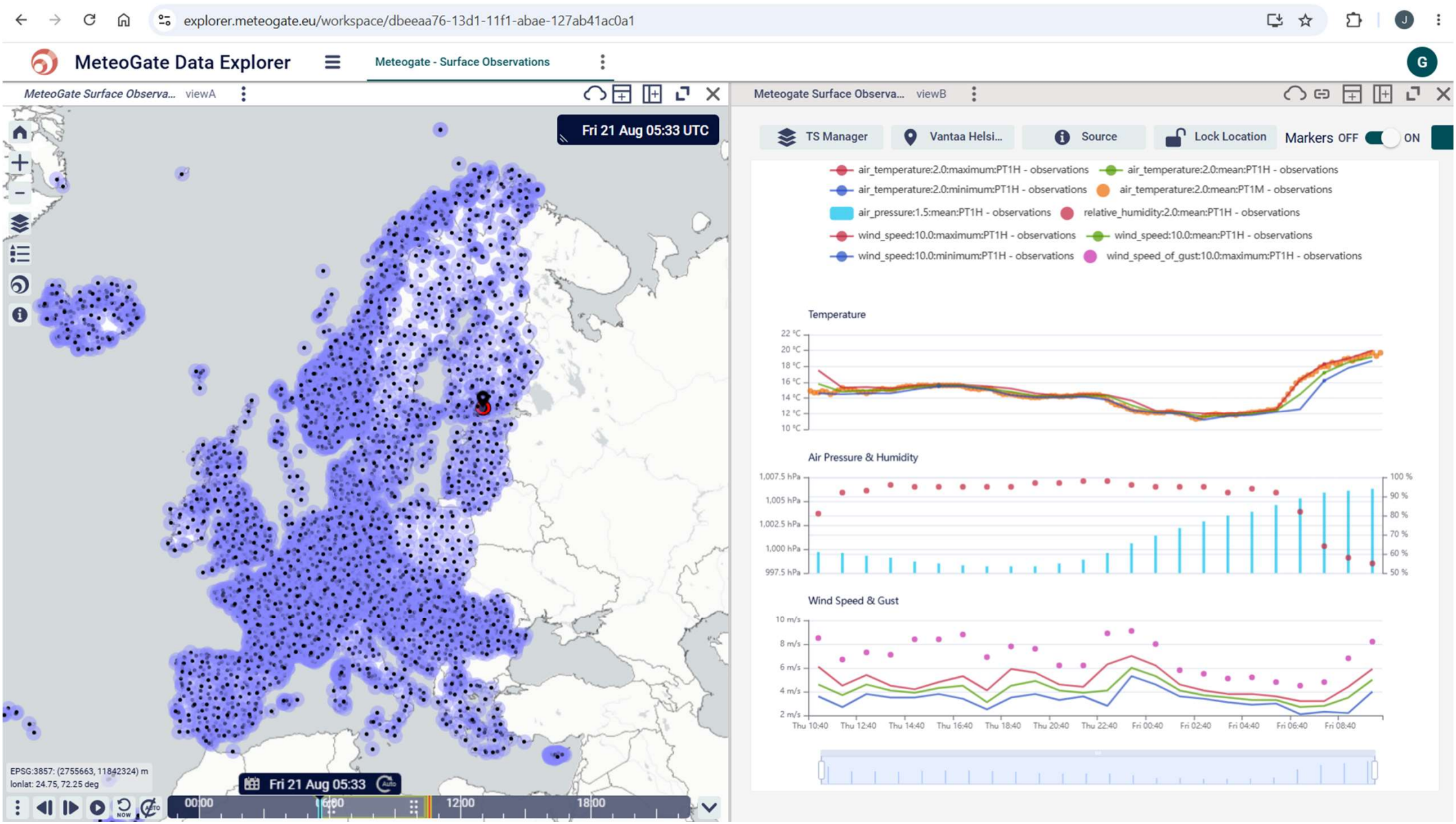
Users can discover what data is available and see if it is suitable before downloading it.

Data Explorer also takes a feed from the World Meteorological Organisation's (WMO) Information System 2.0 (WIS2) catalogue, so other WMO data can be found on it.



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explorer.meteogate.eu/workspace/dbeeaa76-13d1-11f1-abae-127ab41ac0a1
MeteoGate Data Explorer
Meteogate - Surface Observations
MeteoGate Surface Observa... viewA
Fri 21 Aug 05:33 UTC
EPSG:3857: (2755663, 11842324) m
lonlat: 24.75, 72.25 deg
Fri 21 Aug 05:33
00:00
06:00
12:00
18:00
Meteogate Surface Observa... viewB
TS Manager
Vantaa Helsi...
Source
Lock Location
Markers OFF
ON
air_temperature:2.0:maximum:PT1H - observations
air_temperature:2.0:mean:PT1H - observations
air_temperature:2.0:minimum:PT1H - observations
air_temperature:2.0:mean:PT1M - observations
air_pressure:1.5:mean:PT1H - observations
relative_humidity:2.0:mean:PT1H - observations
wind_speed:10.0:maximum:PT1H - observations
wind_speed:10.0:mean:PT1H - observations
wind_speed:10.0:minimum:PT1H - observations
wind_speed_of_gust:10.0:maximum:PT1H - observations
Temperature
22 °C
20 °C
18 °C
16 °C
14 °C
12 °C
10 °C
Air Pressure & Humidity
1,007.5 hPa
1,005 hPa
1,002.5 hPa
1,000 hPa
997.5 hPa
100 %
90 %
80 %
70 %
60 %
50 %
Wind Speed & Gust
10 m/s
8 m/s
6 m/s
4 m/s
2 m/s
Thu 10:40
Thu 12:40
Thu 14:40
Thu 16:40
Thu 18:40
Thu 20:40
Thu 22:40
Fri 00:40
Fri 02:40
Fri 04:40
Fri 06:40
Fri 08:40

MeteoGate

WEBSITE

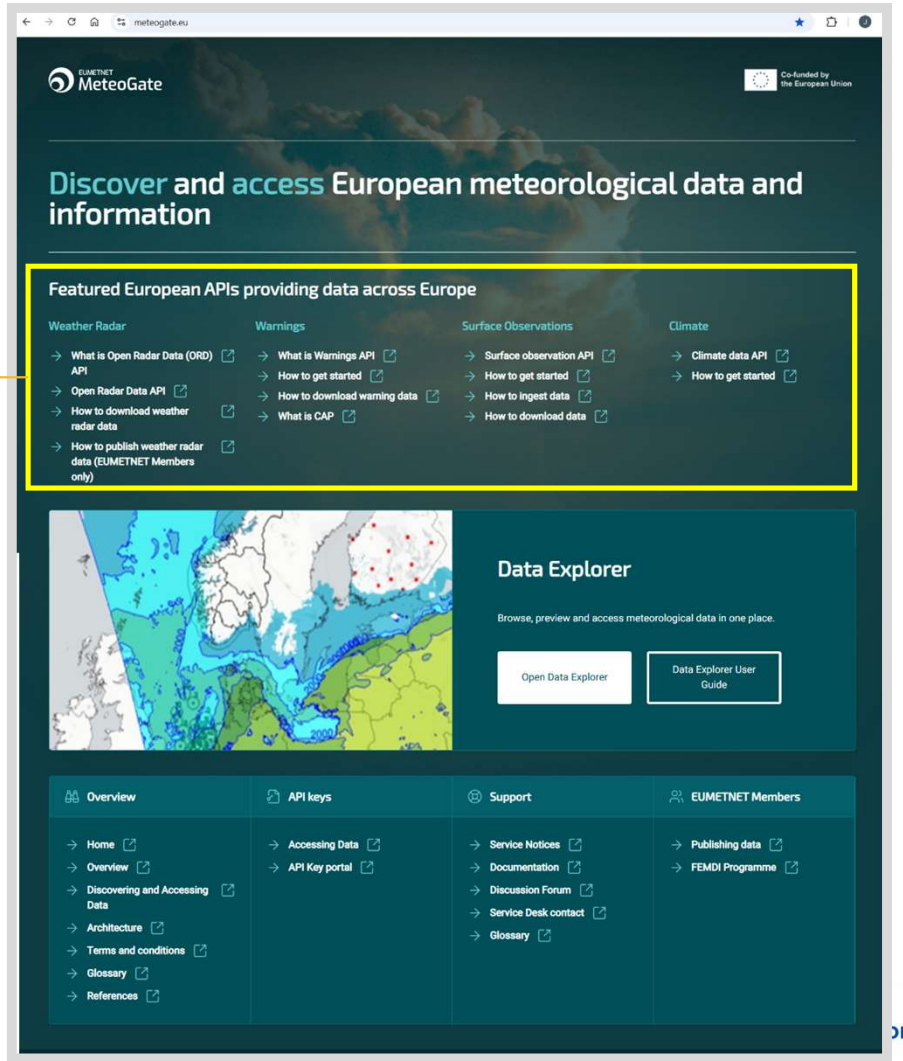
ACCESS DATA

Links to European APIs

EUMETNET have developed and run community data hubs in the data space, with APIs for the following data sets:

- Weather Radar
- Surface Observations
- Warnings
- Climate data

Unlike WMO formats, the APIs use a common, international format, so the data stream is easy to use in applications, and combine with other data.



OASIS

Observations Acquisition Storage & Interface Software

As part of the data space development, software called OASIS has been developed. OASIS is a data-hub engine, used to allow different types of observation data to be ingested, stored, and exposed through a common technical architecture.

EUMETNET use it for their observations, climate and radar data.

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APIs

OBSERVATIONS

Observations Ingest API

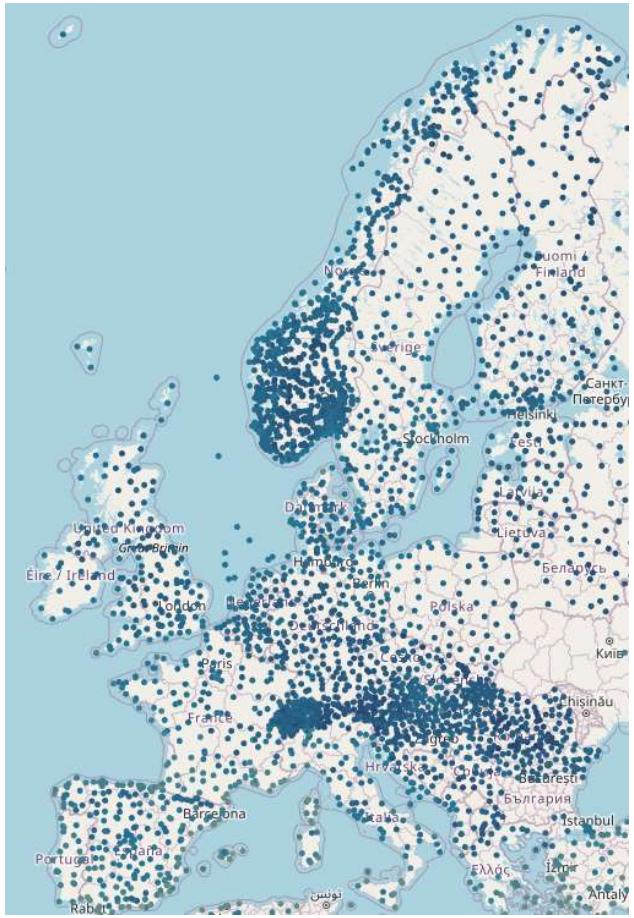
- Developed for easy and structured ingest of observations from NMSs
- Observations ingested as:
  - JSON structure, including required metadata, observations values and links to required vocab services and the parenting dataset hosted by the NMS
  - BUFR files of the same form as those sent to GTS

Observations EDR API

- Uses the Open Geospatial Consortium (OGC) API for Environmental Data Retrieval (OGC-API EDR) interface for retrieving observations from the system

WIS2.0 integration

- Data ingested to E-SOH will be made available for WIS 2.0 through MQTT for real time access. This means that using MeteoGate also makes an NMS compliant with WMO WIS2 requirements



APIs

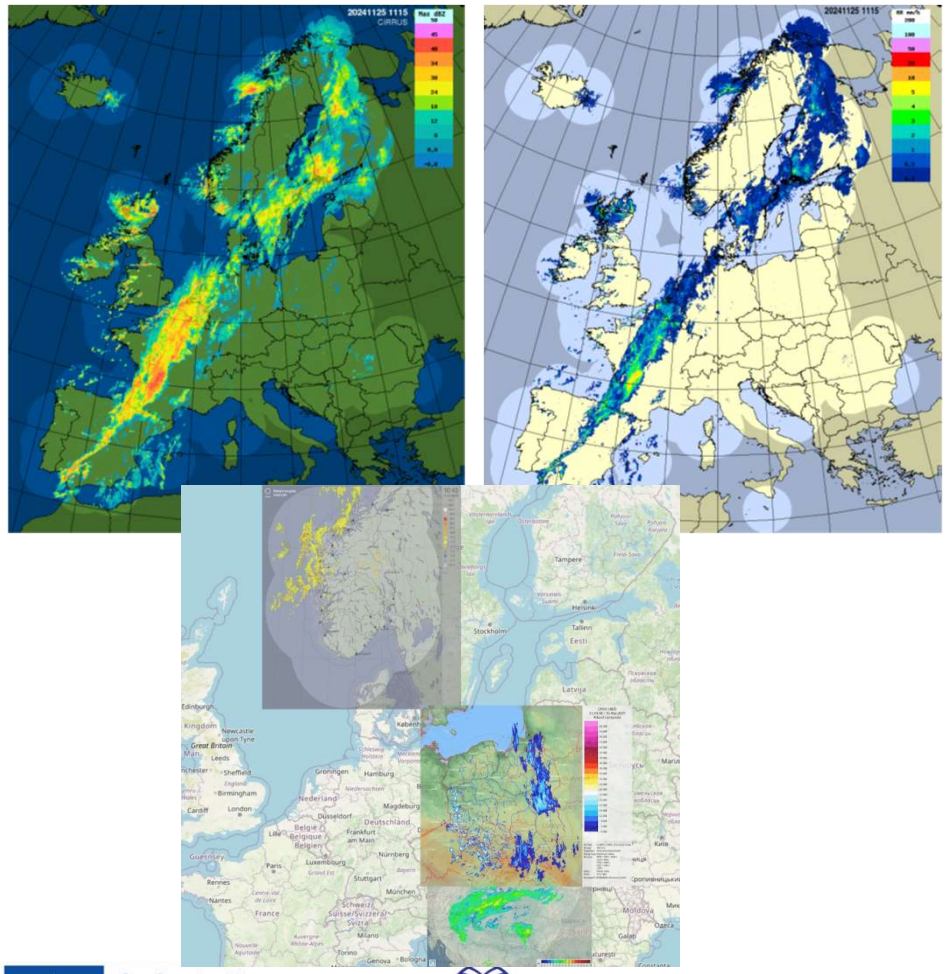
RAINFALL RADAR

Radar Data Ingest API

- Allows EUMETNET to make the European single site radar data and composite products available for downloading.
- Allows NMS to share their products which are stored locally.

Radar Data OGC-API EDR

- EUMETNET stored single site radar data and the composite products data archives supplied using ORD EDR API or direct link to the S3 buckets.
- National datasets stored in local store at an NMS can be made available for downloading via the ORD EDR API.



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APIs

CLIMATE

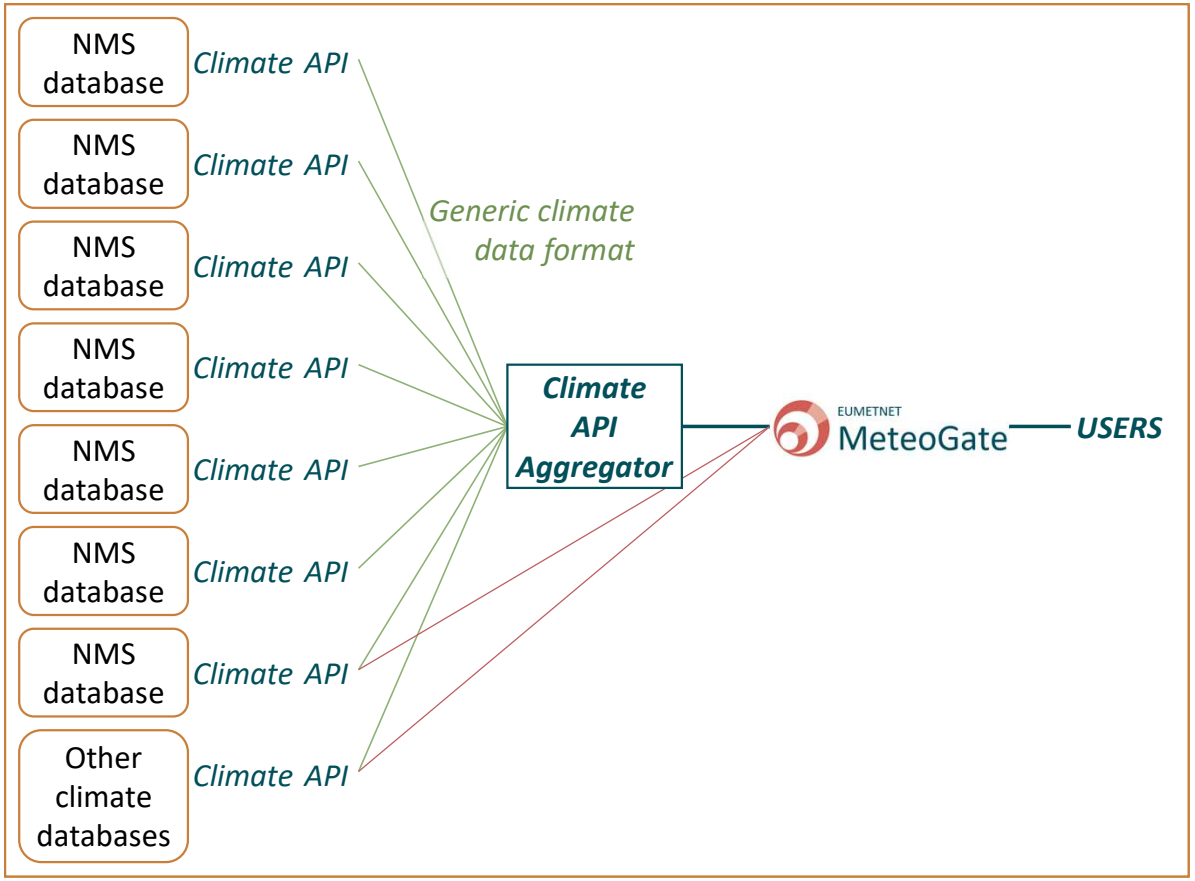
Individual NMS' Climate APIs

- Enables NMSs to share their daily and hourly data using an OGC-API EDR on their in-situ observational climate data.
- This allows data queries in a spatio-temporal manner.

Climate Aggregator

- Wraps climate data from a subset of Essential Climate Variables from individual APIs into one output file (CoverageJSON and NetCDF) in a federated manner.

Users can access individual and aggregated climate datasets.



APIs

METEOALARM WARNINGS

Real-Time Warning API

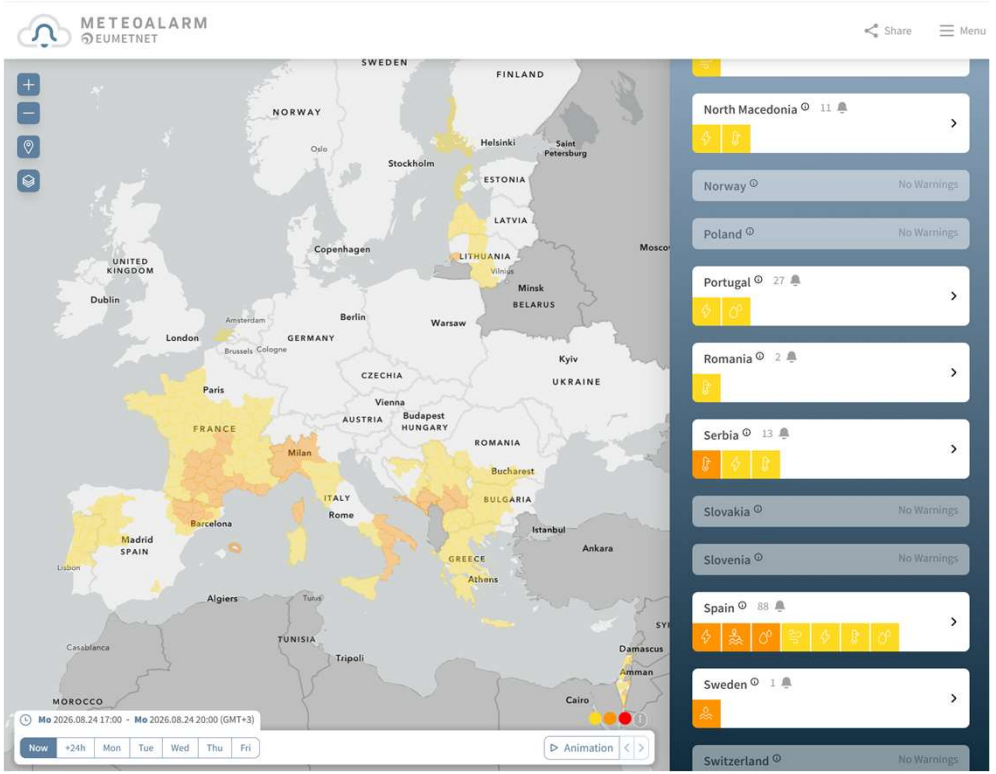
- Access and query ongoing early warnings from 39 NMSs
- Receive notifications of new warnings

Archived Warning API

- Access and query archived early warnings from 39 NMSs

Storm Name API

- Match ongoing early warnings with the official name of the storm, including the necessary backend services, provided that the name is included in the CAP message.
- Users can filter by Country, Awareness Level, and Hazard Type



Major redistributors

- MeteoAlarm information is shared by Google, The Weather Company, AccuWeather, Apple and others.
- Compliance is monitored to ensure quality and recognition of the authoritative NMSs.

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The Weather Company

Google

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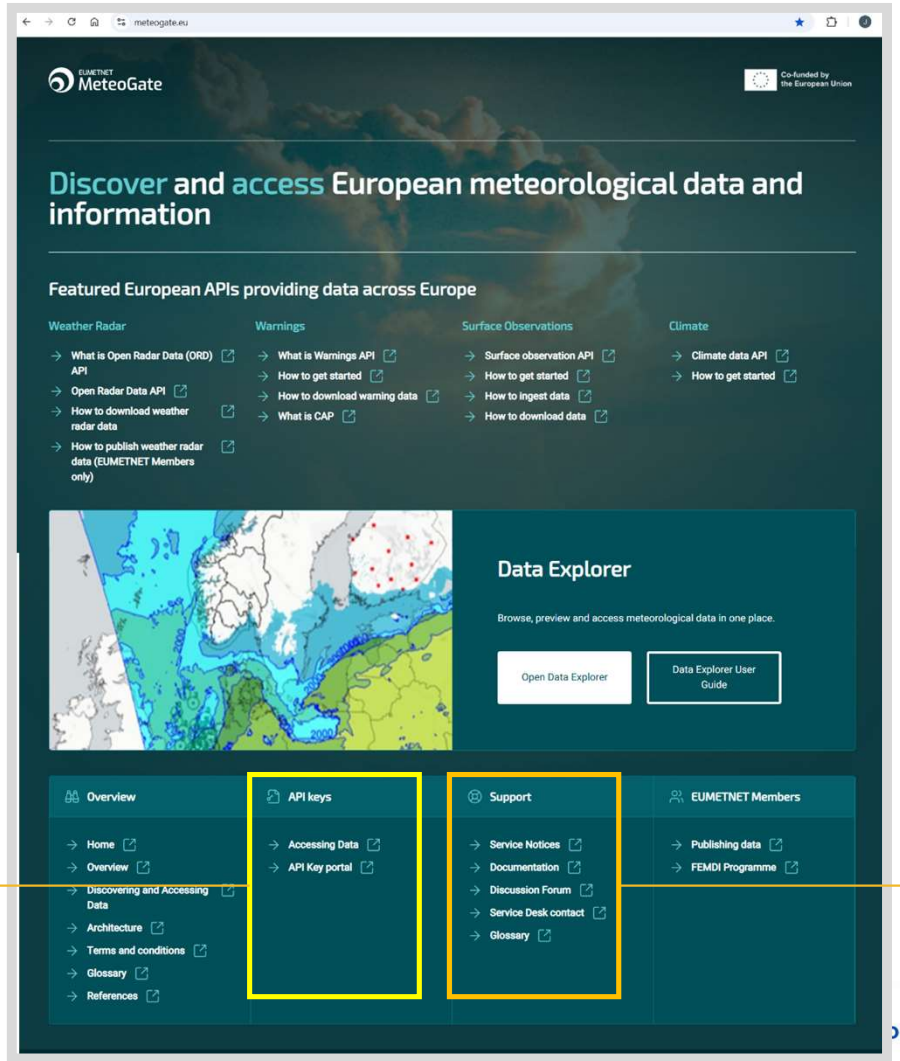
MeteoGate

SINGLE API KEY

REGISTERED USERS
Links to API Key Portal

- Register once for an API key, use with any of the datasets available in MeteoGate
- See available routes
- Get better rate limits

In the first 3 months after going live in July 2026, over 900 Users have registered. This includes Amazon, weathernews, ESRI and ~150 other businesses as well as individuals.



Operated and maintained by FMI

- 24x7 support starting early 2027
- User forum set up
- Lots of documentation available

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MeteoGate

POLICIES

An important part of MeteoGate are the underlying agreements. There are four important policies which have been agreed by all the EUMETMET Members:

- Data Governance
- Incident and Problem Management
- Cyber Security
- Participation Management

EUMETNET have given responsibility to the Finnish Meteorological Institute to operate and maintain the MeteoGate Community Components, and work with the API Data providers.



MeteoGate

BUSINESS BENEFITS

Benefits of MeteoGate and data access using APIs include:

- Increase in exploitation of shared data and data products --> Increase in value to the wider community
- Use of effective standards enables agility and responsiveness --> Improved data sharing
- Improved data sharing between participating Members --> Increased recognition as an authoritative provider of Hydro- Meteorological data
- Improved collection and generation of data and data products --> Less duplication of effort and expenditure
- Improved collaboration and sharing of data and data product --> More affordable and "competitive" product generation
- Improved access and quality of data and data products --> Improved decision making for downstream usage

MeteoGate

BUSINESS BENEFITS

For any further information, please contact Jane Wardle, MeteoGate Programme lead

jane.wardle@fmi.fi

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